Total No. of Questions : 8]

SEAT No. :

PE-897

[Total No. of Pages : 3

[6581]-1903

F.E.

SYSTEMS IN MECHANICAL ENGINEERING

(2019 Pattern) (Semester - I / II) (102003)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right side indicate full marks.*
4) *Use of calculator is allowed.*
5) *Assume suitable data if necessary.*

Q1) a) Classify automobiles based on various considerations. **[7]**

b) Define vehicle specification. Explain following engine specifications **[7]**

i) Torque

ii) Power and

iii) Cubic Capacity

c) Compare vehicle specifications for two-wheeler and three-wheeler vehicles. **[4]**

OR

Q2) a) Explain various components of S. I engine with neat sketch. **[7]**

b) Explain electric vehicle with neat sketch. Mention its components. **[7]**

c) State differences between electric and hybrid vehicle with examples. **[4]**

Q3) a) State importance of suspension system. Explain telescopic suspension system with neat sketch. **[7]**

b) With neat sketch explain the construction and working of disc brake system. **[7]**

c) Why safety arrangements needed in vehicle? Explain the importance of seat belts and air bags in the vehicle. **[3]**

P.T.O.

OR

Q4) a) Define Gear Ratio of gear box. A pinion of pitch circle diameter 150 mm meshes with a gear having 80 teeth. Gear ratio is 4 and speed of gear is 500 rpm. Determine **[7]**

i) Pitch circle diameter of gear

ii) Speed of pinion

iii) Number of teeth on pinion.

b) Explain the working principle of ABS system in vehicle with neat sketch. State its importance over conventional braking system. **[7]**

c) Explain working of water-cooling system in vehicle with neat diagram. **[3]**

Q5) a) With neat sketch explain the shielded metal arc welding. State its applications. **[7]**

b) State significance of Metal Cutting process in industry. Explain following metal cutting processes. **[7]**

i) Turning

ii) Milling and

iii) Drilling operation with neat sketch.

c) Explain concept of Internet of Things (IoT) and its applications in manufacturing. **[4]**

OR

Q6) a) Explain sand casting process with neat sketch. State its advantages and disadvantages. **[7]**

b) State the importance of sheet metal working in manufacturing. Explain Punching and Blanking with neat sketch. **[7]**

c) Draw a block diagram for a 3D printer with all its components. **[4]**

Q7) a) Using block diagram, explain the application of blower in kitchen chimney and vacuum cleaner. **[7]**

b) Explain with neat sketch working of water pump used for lifting water for overhead tank. **[7]**

c) An electric motor driven pump fills an over headed tank placed at a height of 20 m from the ground level. The mass of the water pumped per second is 5.56kg. Input power of the motor is 2200 W. Calculate the efficiency of the motor. (use $g = 9.81\ m/s^2$) **[3]**

OR

Q8) a) With the help of block diagram, explain working of electric geyser. State various specifications for an electric geyser. **[7]**

b) Explain with neat sketch working of refrigerator. State its domestic and industrial applications. **[7]**

c) A refrigerator has working temperatures in the evaporator and condenser coils as -30^0C and 32^0C. What is the maximum COP of the system? Draw its block diagram. **[3]**

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